# RNA-Seq Gene Expression for Cancer Type Prediction

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| Goal | Dataset Scale | Best Result |
|---|---|---|
| **Cancer-type prediction**<br>Use gene-expression patterns to classify cancer subtypes and cluster uncertain samples. | **20,531 features**<br>Each patient record contains 20,531 RNA expression values before dimensionality reduction. | **92.5% accuracy**<br>Top hierarchical settings (ward / average) outperformed most alternatives; K-Means reached 91.6%. |

## Overview

This project investigated cancer type prediction using RNA-Seq gene expression data, with a particular focus on identifying samples from unknown classes. I combined ***SVM classification*** with ***K-Means*** and ***Hierarchical Clustering***, allowing low-confidence samples to be further analyzed rather than forced into known categories. The workflow also included ***KNN imputation***, ***PCA-based dimensionality reduction***, and class balancing with ***KMeans-SMOTE*** and ***ADASYN***.
Results showed that the proposed hybrid approach improved class separability, with the best clustering configuration achieving ***92.5%*** accuracy.

## Methodological Pipeline

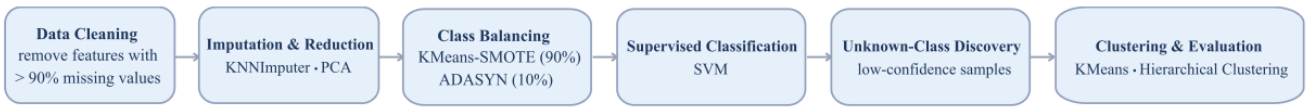


*Figure 1. Proposed analytical pipeline for cancer type prediction: preprocessing and rebalancing were followed by SVM classification, while low-confidence samples (confidence < 0.95) were further analyzed using clustering.*

**1. Data cleaning:** The dataset originally contained ***20,531 gene expression features*** with a missing rate of ***14.1%***. By removing features with more than ***90% missing values***, the missing rate was reduced to ***0.4%***, while feature dimensionality decreased by ***22.6%***.

**2. Imputation & Dimensionality reduction:** The remaining missing values were completed using ***KNNImputer***. To address the curse of dimensionality, ***PCA*** was applied to reduce the feature space from ***20531*** dimensions to ***292 principal components***, while remaining ***95%*** of the total variance. This resulted in an overall dimensionality reduction of approximately ***98.6%***.

**3. Class balancing (Figure 2):** Because the dataset was highly imbalanced across cancer types, a hybrid oversampling strategy was adopted using ***KMeans-SMOTE (90%)*** and ***ADASYN (10%)***. This combination improved class balance while also strengthening samples near difficult decision boundaries.

**4. SVM-based classification:** ***SVM*** was selected as the primary classifier due to its strong performance on high-dimensional data. In internal validation on known classes, the model demonstrated very strong classification performance, and a confidence threshold of ***0.95*** was applied to identify uncertain predictions.

**5. Unknown class discovery:** Samples with confidence scores below ***0.95*** were routed to unsupervised clustering for further analysis instead of being directly assigned to known categories. This design allowed the system to better handle potential ***unknown cancer classes*** outside the training set.

**6. Clustering and Evaluation:** Two clustering methods were compared: ***K-Means*** and ***Hierarchical Clustering***. The results showed that ***K-Means reached 91.6% accuracy***, while ***Hierarchical Clustering achieved up to 92.5%***, though performance varied significantly across linkage strategies.

## Post-PCA Class Distribution After Rebalancing

### PCA and rebalancing improved class separability.

Figure 2 shows clearer class structure after dimensionality reduction and hybrid oversampling, indicating that the preprocessing pipeline enhanced the quality of the feature space.

### Hybrid rebalancing strengthened performance.

The combination of ***KMeans-SMOTE (90%)*** and ***ADASYN (10%)*** improved class balance while preserving harder-to-learn samples near decision boundaries.

### Cluster selection was not based on a single metric.

Both ***Silhouette Analysis*** and the ***Elbow Method*** were used to determine cluster counts before comparing clustering accuracy across settings.

*Figure 2. Post-PCA class distribution after hybrid rebalancing with KMeans_SMOTE (90%) + ADASYN (10%).*

## Clustering Performance by Method and Cluster Selection Criteria

| Method | Clusters (Silhouette) | Accuracy | Clusters (Elbow) | Accuracy |
|---|---|---|---|---|
| K-Means | 2 | 91.6% | 2 | 91.6% |
| Hierarchical - single | 2 | 53.3% | 9 | 53.3% |
| Hierarchical - complete | 2 | 92.2% | 2 | 92.2% |
| Hierarchical - average | 10 | 90.4% | 6 | 92.5% |
| Hierarchical - ward | 2 | 92.5% | 2 | 92.5% |

*Table 1. Most hierarchical-linkage settings exceeded 90% accuracy, while single-linkage performed substantially worse. K-Means achieved 91.6% accuracy.*

## Technical Takeaways and Relevance to Fintech

| Key Insight | Relevance to Fintech |
|---|---|
| Clustering performance varied substantially by linkage strategy: hierarchical methods achieved up to ***92.5% accuracy***, whereas single linkage dropped to ***53.3%***. | This showed that methodological choices can materially affect downstream outcomes, reinforcing the importance of model selection, distance metrics, and comparative evaluation in data-driven systems such as ***risk segmentation*** and ***anomaly detection***. |
| The project required handling ***high-dimensional, incomplete, and imbalanced data*** through cleaning, imputation, PCA, and hybrid oversampling before reliable modeling could be achieved. | This experience is directly transferable to FinTech, where real-world datasets are often noisy and unevenly distributed, particularly in applications such as ***fraud detection, credit risk modeling, and transaction behavior analysis.*** |